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Oefeningenreeks 2D nr. 14.9

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\sin(ax)}{\tan(bx)} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin(ax) \cdot ax}{ax}}{\frac{\tan(bx) \cdot bx}{bx}} \\ &= \frac{\lim_{ax \rightarrow 0} \frac{\sin(ax)}{ax}}{\lim_{bx \rightarrow 0} \frac{\tan(bx)}{bx}} \cdot \lim_{x \rightarrow 0} \frac{ax}{bx} \\ &= \frac{1}{1} \cdot \lim_{x \rightarrow 0} \frac{a}{b} = 1 \cdot \frac{a}{b} = \frac{a}{b} \end{aligned}$$

References